



JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER & SYSTEMS SCIENCES
NEW DELHI-110067

MID – SEMESTER (2024)

Class: MCA III Semester

Course Title: Wireless Networks

Course Code: CS446

1. Give application where contention-based, reservation-based and scheduling based MAC protocols can be used and also write pros and cons one over another. (3)
2. High-frequency X-rays and Gamma rays are not normally used for wireless communication. Explain why (2)
3. Write the difference among 1G,2G,3G, 4G and 5G. (5)
4. Explain the Radio Propagation Mechanism. (5)
5. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10⁸ bits per second) over a 1 km (kilometre) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable? (3)
6. A pure ALOHA network transmits 200 – bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision free? (2)
7. What are the seven layers of the OSI model and explain their basic functions of layers? (5)
8. Explain free space and 2 ray path loss model? (3)
9. What is CSMA/CA and explain the variants of CSMA /CA. (2)

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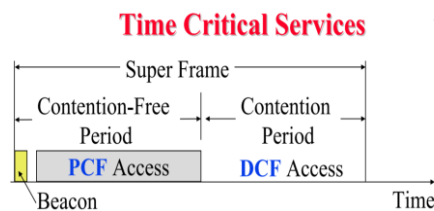
Duration: 3:00 Hours
Max Marks: 70

Q1. What do you mean by Ad Hoc networks? Why Ad-Hoc Networks are needed? [5]

Q2. Write the difference between ad-hoc wireless networks and cellular networks? (at least 10 difference). [5]

Q3. Explain in following in detail about IEEE 802.11:

- a. IEEE 802.11 features. [3]
- b. IEEE 802.11 physical layers. [3]
- c. Explain the CSMA/CA with functional diagram. [5]
- d. Describe each fields of IEEE 802.11 architecture with diagram. [5]
- e. Frame Format and explain each field. [5]
- f. Describe all the priorities of IEEE 802.11 and also sketch the diagram? [5]
(Priorities: *IFS, SIFS, DIFS and PIFS*) .
- g. Write detail note about both time critical services PCF and DCF and also explain with diagram. [4]



- h. Power Management IEEE 802.11. [2]
- i. Pros and Cons of 802.11 (at least 5 points of both). [4]

Q4. What is the maximum data rate that can be supported on a 10 MHz noiseless channel, if the channel uses eight-level digital signals? [2]

Q5. Calculate the maximum achievable data rate over a 9 KHz channel whose signal to noise ratio is 20 dB. [2]

Q6. Explain in following in detail: [3+3+3+3+3]

- a. Small scale fading and Large scale fading.
- b. Nyquist Theorem and Shannon Theorem.
- c. Doppler Shift and Path Loss.
- d. Adjacent Channel interference and Co-Channel Interference.
- e. Explain three versions of CSMA.

Q7. Explain and classify the multiple access protocols. [5]